



E0704

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Maschinelles Lernen

Exam: In 2064 / Endterm

Date: Friday 21st February, 2025

Examiner: Prof. Dr. Stefan Leutenegger
Dr. Leo Schwinn

Time: 16:30 – 18:30

P 1 P 2 P 3 P 4 P 5 P 6 P 7 P 8 P 9

I								
---	--	--	--	--	--	--	--	--

Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 40 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____



☐ Exam empty





Problem 1 Decision Trees and KNN (3 credits)

0

1

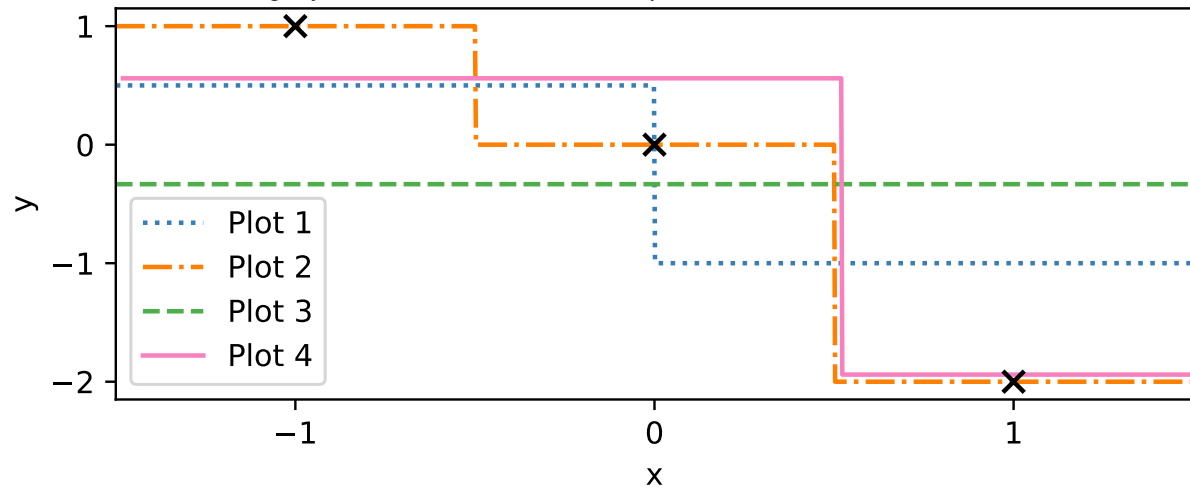
2

3

Plotted below is a dataset of three points $\{(x_i, y_i)^T \in \mathbb{R}^2 \mid i \in \{1, 2, 3\}\}$, each of them marked by an X. The task is to regress y label over the x feature.

For each of the regressors (a) - (e) listed below identify which plotted line corresponds to its optimally-fitted prediction. Multiple regressors can have same optimally-fitted plots. Briefly **justify** each answer.

The line of Plot 4 is slightly offset to avoid visual overlap with Plots 1 and 2.



- (a) 1-NN regressor with L_1 distance measure and uniform weight function
- (b) 2-NN regressor with L_∞ distance measure and uniform weight function
- (c) 3-NN regressor with L_1 distance measure and uniform weight function
- (d) Decision tree regressor with maximum depth 2 (i.e. two splits) and squared error impurity function
- (e) Decision tree regressor with maximum depth 1 (i.e. one split) and squared error impurity function





Problem 2 Probabilistic Inference and Linear Regression (5 credits)

Consider the following probabilistic interpretation of a regression model:

$$p(y|x; \mathbf{w}, b) = \frac{\beta}{2\alpha\Gamma(1/\beta)} e^{-\left(\frac{y - \mathbf{x}^\top \mathbf{w} - b}{\alpha}\right)^\beta} \quad (2.1)$$

with input vector $\mathbf{x} \in \mathbb{R}^D$, scalar target $y \in \mathbb{R}$, and parameters $\mathbf{w} \in \mathbb{R}^D$ as well as $b \in \mathbb{R}$. $\alpha > 0$ is a positive real number and β is a positive natural even number ($\beta \in \{2, 4, 6, \dots\}$). $\Gamma(z)$ is the gamma function $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$.

a) **Derive** a quantity $g(\mathcal{D}|\mathbf{w}, b)$ that is proportional to the likelihood $g(\mathcal{D}|\mathbf{w}, b) \propto p(\mathcal{D}|\mathbf{w}, b)$ w.r.t. $\mathbf{w}$ and b for an i.i.d. dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$.



b) Since the maximum likelihood estimate (MLE) does not allow for a simple closed-form solution for all possible α and β we will optimize the parameters using gradient ascent with step size $\eta > 0$.

Derive the parameter update for step t that yields $\mathbf{w}_t, b_t$ with regard to the previous parameters $\mathbf{w}_{t-1}, b_{t-1}$. *Note: for simplicity consider the log of the quantity $g(\mathcal{D}|\mathbf{w}, b)$ that you derived in the previous problem, $\log g(\mathcal{D}|\mathbf{w}, b)$.*





- 0
- 1
- 2

c) *Maximum a posteriori estimate (MAP)*: **derive** an adapted gradient ascent update for b_t as a function of b_{t-1} to incorporate the prior $p(b) = \frac{\delta}{2\gamma\Gamma(1/\delta)} e^{-\left(\frac{|b|}{\gamma}\right)^\delta}$ with $\gamma > 0$ and $\delta > 0$.

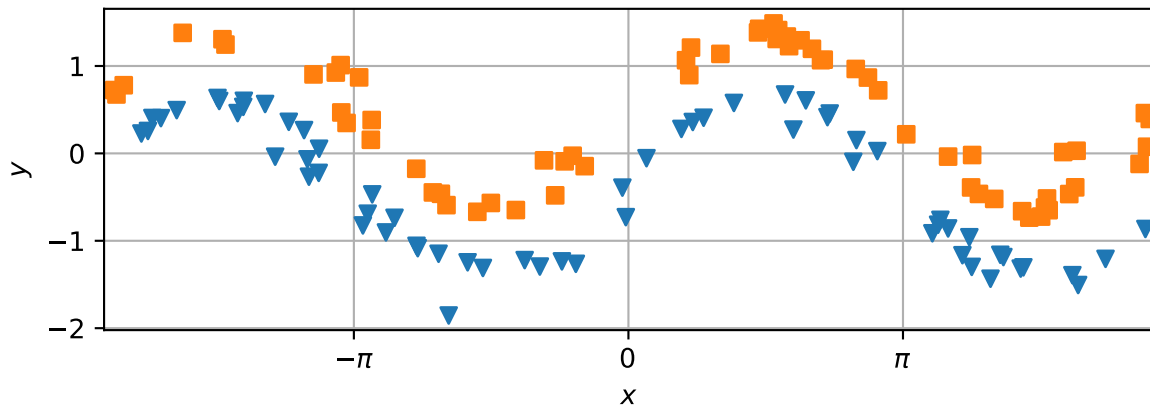




Problem 3 Linear Classification (5 credits)

a) You have a generative model $p(\mathbf{x} | y)$ for $y \in \{1, 2, 3\}$. Assume that all classes are equally likely a priori. How can you use this model to classify a new data point $\mathbf{x}$? **Justify** your answer by describing a procedure $f(\mathbf{x})$ that returns the estimated class.

0
1
2



b) Can a linear classifier distinguish between the orange boxes and the blue triangles in the two-dimensional dataset above? If yes, **construct** such a classifier explicitly, i.e. write down a fully specified function that returns 1 for an orange box and 0 for a blue triangle or vice versa. If no, **describe** why no linear classifier can solve this problem.

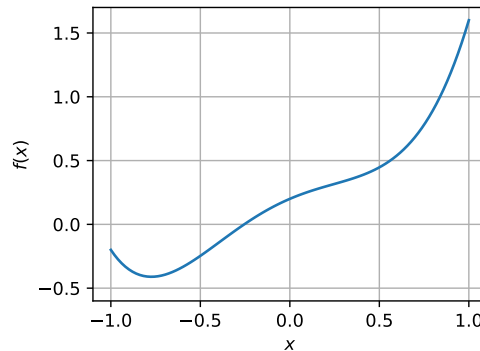
0
1
2
3



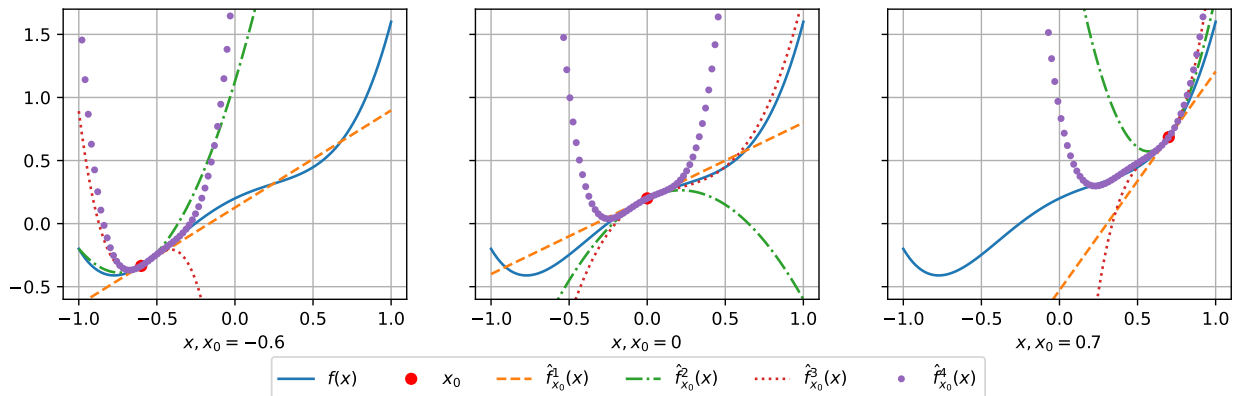


Problem 4 Optimization (6 credits)

You are given the function $f : [-1, 1] \rightarrow \mathbb{R}$ as plotted here



To find its minimum, you decide to use iterative m -order Taylor expansions $\hat{f}_{x_0}^m : [-1, 1] \rightarrow \mathbb{R}$ starting from some point $x_0 \in [-1, 1]$. You plot $\hat{f}_{x_0}^m$ for different $m \in \{1, 2, 3, 4\}$ and $x_0 \in \{-0.6, 0, 0.7\}$ in the following plot:



- 0 ☐ a) You decide to use Newton's method to find the minimum of f . Is it guaranteed that the function value decreases after every iteration, i.e., $f(x_{t+1}) \leq f(x_t)$? **Justify** your answer using the insights from the plots above.

1 ☐

2 ☐

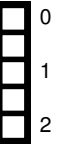
- 0 ☐ b) **Derive** the optimal update step using the 2nd Taylor expansion at $x_0 = -0.6$. **Verify** that the optimal update minimizes the 2nd Taylor expansion at $x_0 = -0.6$. You already computed the 2nd-order Taylor expansion $\hat{f}_{-0.6}^2$:

1 ☐

2 ☐

$$\hat{f}_{-0.6}^2(x) = -0.2 + (x + 0.6) + 1.3(x + 0.6)^2. \quad (4.1)$$





c) Why are >2 degree optimization techniques very rarely used for functions $f : \mathbb{R}^D \rightarrow \mathbb{R}$ with $D > 1$? State at least two reasons with **justification**.





Problem 5 Deep Learning (4 credits)

You are implementing a deep learning model to be trained with backpropagation. Somewhere in your model you have the formula

$$\mathbf{z} = \mathbf{a} \cdot \mathbf{b} \cdot \mathbf{c} \quad (5.1)$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^N$ are some inputs and $\mathbf{z} \in \mathbb{R}^N$ is the output of the operation. Broadcasting applies in the above formula as usual in numpy and pytorch code.

For your implementation, you have access to a function `mul(a, b)` that takes two tensors and multiplies them elementwise. `mul` has a backward pass implemented for automatic differentiation in a straightforward way.

0 ☐
1 ☐

a) **Draw** a computational graph corresponding to eq. (5.1) based on `mul`.

0 ☐
1 ☐

b) How much excess memory do you need for eq. (5.1) between the forward and backward pass during training? Excess memory refers to any memory on top of the memory required to store $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{z}$. Answer in bits. All numbers are represented as 32-bit floats. **Justify** your answer.

You look through the documentation of your deep learning library and discover the obscure function `mulmul(a, b, c)` that multiplies three numbers together in one go.

0 ☐
1 ☐

c) **Draw** a computational graph corresponding to eq. (5.1) based on `mulmul`.





☐ 0
☐ 1

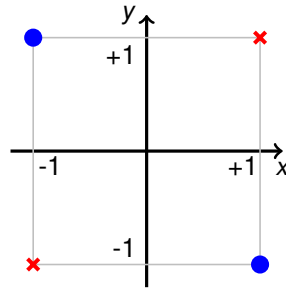
d) What is the minimum amount of excess memory required between forward and backward pass if you use `mulmul` instead of `mul`? Answer *in bits*. All numbers are represented as 32-bit floats. **Justify** your answer by describing how the backward pass of `mulmul` could be implemented.





Problem 6 SVM and Kernels (4 credits)

Below you see a dataset of four $\mathbb{R}^2$ points on the corners of a centered length-2 square, coming from two classes marked by dot vs. cross markers.



- 0 ☐ a) Suppose you want to solve the hard-margin SVM dual problem with linear kernel on the dataset above. Is the primal problem feasible? What behaviour do you expect in the dual problem from the Lagrange multipliers? **Justify** your answer.
- 1 ☐

- 0 ☐ b) Write down a kernel function $k: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ for which the hard-margin SVM problem is better-behaved than in subtask (a). Explicitly **verify** that your k is indeed a kernel. You may use all the rules for kernel-preserving operations discussed in the lecture.
- 1 ☐ *Hint:* You may try first finding a finite-dimensional feature map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^n$, $n \in \mathbb{N}$, such that the SVM problem becomes more well-behaved in the image space of your feature map.
- 2 ☐





c) Assume that you want to keep the linear kernel and also leave the data unmodified. What other design choice could you make to fix the behavior from subtask (a)? Would this likely lead to a good classifier? **Justify** your answer.





Problem 7 Dimensionality reduction (6 credits)

Assume we are given a data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$. Further, we know that $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}^T \mathbf{X}) = k$.

- 0 ☐
1 ☐
2 ☐
- a) If we project $\mathbf{X}$ onto the subspace spanned by its first k principal components, can we reconstruct $\mathbf{X}$ without any loss? **Justify** your answer.
Hint: You are allowed to use any finding of the two PCA derivations.

- 0 ☐
1 ☐
- b) Does dimensionality reduction of $\mathbf{X}$ using the **first** k singular vectors always lead to a loss-less compression? **Justify** your answer based on the singular values.

- 0 ☐
1 ☐
- c) Consider an Autoencoder with a latent space of dimension $m > k$, where latent variables are ordered by their expected variance after projection. Does dimensionality reduction of $\mathbf{X}$ using the **first** k dimensions of the latent variable always lead to a loss-less compression? **Justify** your answer.

- 0 ☐
1 ☐
2 ☐
- d) Given a linear regression problem with target $y \in \mathbb{R}$, assume that $k < n, d$. Can SVD or PCA help us find the optimal parameters in closed form? **Justify** your answer.





Problem 8 Eigenvector Computation (Dimensionality Reduction) (5 credits)

You want to compute the first eigenvector of a diagonalizable matrix $\mathbf{A}$ by iterating

$$\mathbf{x}^{(k+1)} = \frac{\mathbf{A}\mathbf{x}^{(k)}}{\|\mathbf{A}\mathbf{x}^{(k)}\|} \quad (8.1)$$

where $\mathbf{x}^{(0)}$ is a randomly initialized vector.

a) **Prove** that the k^{th} iteration can be written as: $\mathbf{x}^{(k)} = \frac{\mathbf{A}^k \mathbf{x}^{(0)}}{\|\mathbf{A}^k \mathbf{x}^{(0)}\|}$.

0
1

b) Assume that $\mathbf{A}$ has a unique dominant eigenvalue, i.e., $|\lambda_1| > |\lambda_2|$. **Prove** that the method converges to $\mathbf{v}_1$ for large k , where $\mathbf{v}_1$ is the eigenvector corresponding to the dominant eigenvalue (i.e., the largest eigenvalue in absolute value). You can directly use the results from a).

0
1
2

Hint: Represent $\mathbf{x}^{(0)}$ in the eigenbasis of $\mathbf{A}$ as $\mathbf{x}^{(0)} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$, where $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are eigenvectors of $\mathbf{A}$.

c) Which factor influences the convergence speed of the method?

0
1





0 ☐
1 ☐

d) Given $\mathbf{v}_1$, how can you compute the corresponding eigenvalue?



2882-01-0013-E0704-14



2882-01-0013-E0704-14



2882-01-0013-E0704-14



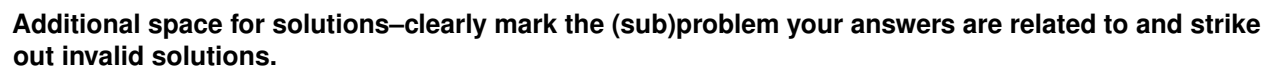


Problem 9 Clustering (2 credits)

What are two advantages of the Gaussian Mixture Model over K-Means clustering?

☐	0
☐	1
☐	2



This image shows a full page of blank graph paper. The grid consists of small, equal-sized squares formed by thin gray lines. There are no margins, text, or other markings on the page.

2882-01-0013-E0704-16